

# Gravitational Fields

## The Gravitational Field

A gravitational field is a field of a force which is a region in space where a particle (mass) experiences a force. define gravitational field as force per unit mass

Assumption:

- For an object outside uniform sphere, the sphere behaves as if its mass is concentrated at its center.



### Newton's Law of Gravitation

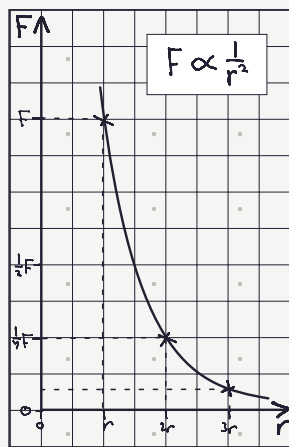
- Law of gravitational force is:

- ① directly proportional to product of masses
- ② Inversely proportional to the square of their separation

- Equation:  $F = \frac{GMm}{r^2}$

- $F$ : gravitational force
- $M$  and  $m$ : masses, where  $M$  is dominant
- $r$ : distance between the masses centers (separation)
- $G$ : gravitational constant ( $6.67 \cdot 10^{-11}$ )

Define gravitational force as force of attraction between masses



### Representing Fields & Field lines

- Lines of gravitational force show the direction of force on a mass

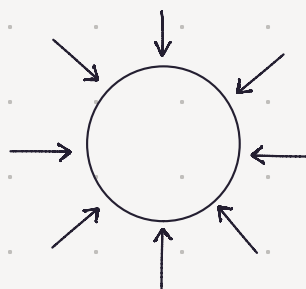
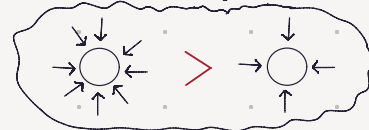
- Field lines show:  
① Direction  
② Strength

- Field patterns:

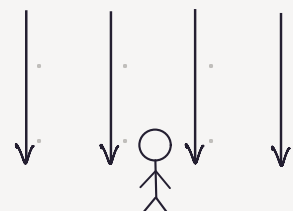
- Radial: Spacing increases with distance  $\rightarrow$  Field strength decreases
- Uniform (Earth surface): Radius of earth is very large compared to height  $\rightarrow$  equal spacing  $\rightarrow$  approximately parallel  $\rightarrow$  constant field strength [More on that later]

- Rules:

- Point toward mass
- Direction of field lines show the direction of gravitational force
- Lines must be equidistant
- Stronger field  $\rightarrow$  less spacing b/w lines, e.g.:



Earth - Radial



Near surface - Uniform

## Gravitational Field Strength

The gravitational field strength at a point is the gravitational force per unit mass

- Unit:  $N kg^{-1}$  or  $ms^{-2}$

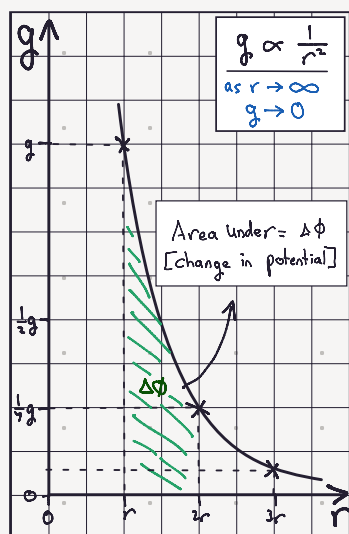
- Equation:  $g = \frac{GM}{r^2}$

- Derivation:

$$g = \frac{F}{m} \quad \text{and} \quad F = \frac{GMm}{r^2}$$

$$\rightarrow g = \frac{\left(\frac{GMm}{r^2}\right)}{m} \rightarrow g = \frac{GM}{r^2}$$

This means that "g" only depends on  $M$  (dominant) not test mass



Why is  $g$  approximately constant near Earth's surface?

— The lines are radial, but near surface the earth's radius is very large compared to height so lines are approximately parallel  $\rightarrow$  constant field strength near surface

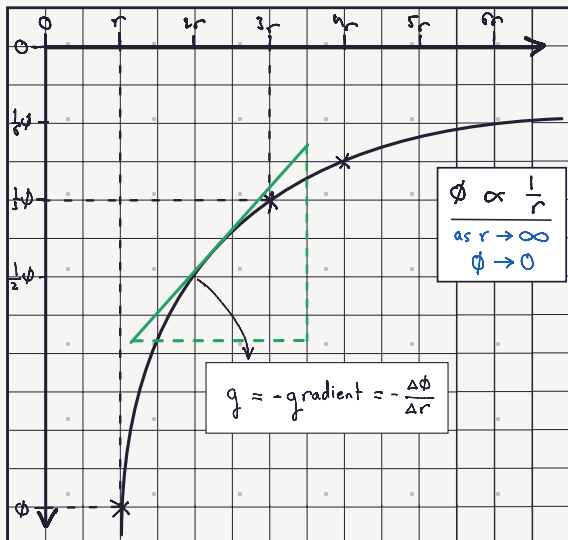
Why is  $g$  numerically equal to gravitational acceleration [of free fall]?

— Both are equal to  $\frac{F}{m}$ ,  
 $- F = ma$   
 $- F = mg$

# Gravitational Potential and Gravitational Potential Energy

**Gravitational Potential Energy ( $E_p$ )**: Work done to bring mass from infinity to the point

- Equation:  $E_p = -\frac{GMm}{r}$       - Unit: J or Nm



**Gravitational Potential**: Work done per unit mass to bring mass from infinity to the point

- Equation:  $\phi = \frac{\text{work done}}{m} = \frac{(-\frac{GMm}{r})}{m} = -\frac{GM}{r}$

- Gravitational potential is independent of mass "m" affected by field

- Unit: J kg<sup>-1</sup>

- Why is  $\phi$  always negative?

Potential is Zero at infinity  $\rightarrow$  Force is attractive

$\rightarrow$  So as mass approaches source  $\rightarrow$  the potential energy decreases

## Escape Velocity

- This is the minimum speed required for an object to completely escape a planet's gravitational field. This means:

- Reaching infinity
- Zero K.E left
- Doesn't fall back

- The object of mass  $m$  has  $E = K.E + P.E = \frac{1}{2}mv^2 - \frac{GMm}{r}$

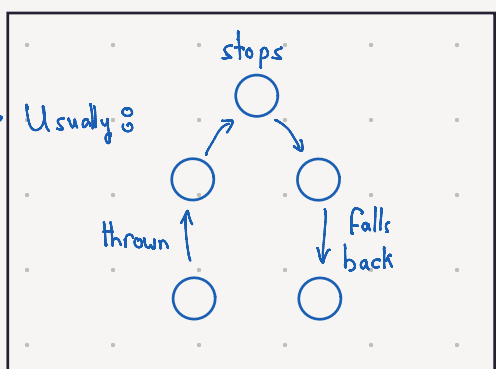
- For an object to escape  $\Delta K.E = -\Delta E_p$ :

$$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -\left[-\frac{GMm}{r_f} - \left(-\frac{GMm}{r_i}\right)\right]$$

$\rightarrow$  Both final K.E. and  $E_p = 0$

$$\rightarrow \text{So, } -\frac{1}{2}mv^2 = -\frac{GMm}{r}$$

$$\rightarrow v = \sqrt{\frac{2GM}{r}}$$



- Key Points:

- Any speed  $\geq v \rightarrow$  escapes
- Only depends on  $M$ , not mass of escaping object "m"

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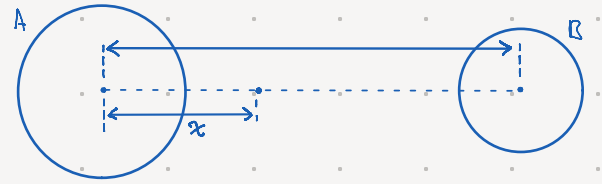
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## Combined Gravitational Fields

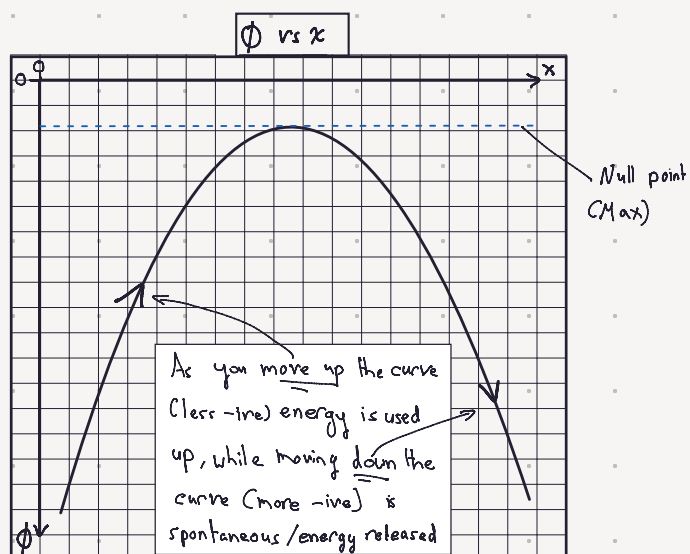
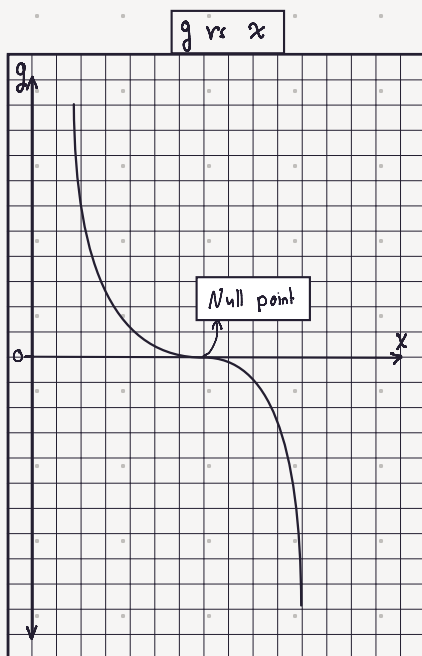
When two or more masses create gravitational fields,

the total field strength at a point is the vector sum of individual field strength:  $g_t = g_1 + g_2 + \dots$  [direction dependent]

However, gravitational potential is a scalar, so total potential is algebraic sum:  $\phi_t = \phi_1 + \phi_2 + \dots$  [direction independent]



$$g_{net} = g_A - g_B \quad \phi_{net} = \phi_A + \phi_B$$



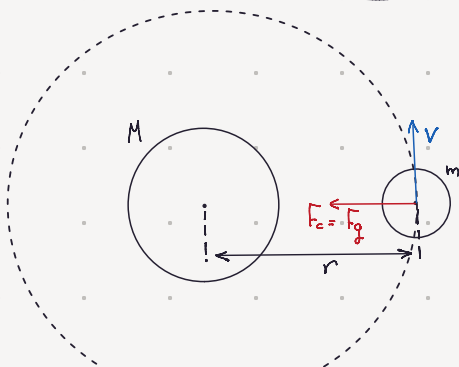
### Null Point

- Point where gravitational fields cancel out [ $g_t = 0$ ], and the gravitational potential is at its max

- Position is not necessarily in the middle, but is closer to smaller mass

## Orbital Motion

For a satellite in a circular orbit, the gravitational force provides the centripetal force:  $F_c = F_g$



### Orbital Speed

- Derivation:  $F_c = F_g \rightarrow \frac{mv^2}{r} = \frac{GMm}{r^2}$   
 $\rightarrow v^2 = \frac{GM}{r} \rightarrow v = \sqrt{\frac{GM}{r}}$

- Key Points:

□ Only depends on M and r, not m

□ Notice that  $v_{escape} = \sqrt{2} \cdot v_{orbital}$

Don't Confuse Them

### Geostationary Satellites

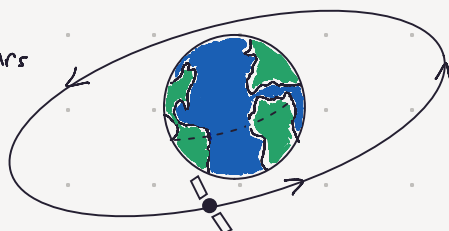
- A satellite that remains above the same point all times

- Features:

□ Has an orbital period of 24 hours

□ Orbits from west to east

□ Orbits directly above equator



### Orbital Period

- Derivation:  $v^2 = \frac{GM}{r}$  and  $v = \frac{2\pi r}{T}$  <sup>Chapter 12</sup>

$\rightarrow \frac{4\pi^2 r^3}{T^2} = \frac{GM}{r} \rightarrow \frac{T^2}{4\pi^2 r^3} = \frac{r}{GM}$

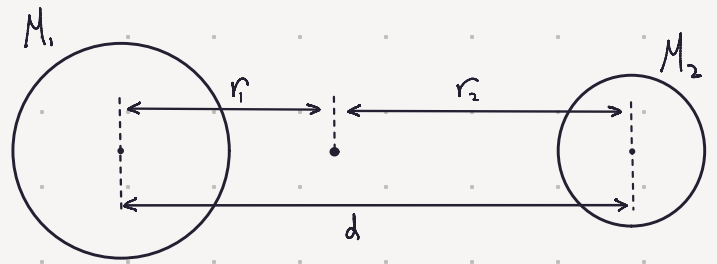
$\rightarrow T^2 = \frac{4\pi^2 r^3}{GM} \rightarrow T = 2\pi \sqrt{\frac{r^3}{GM}}$

- Key Points:

□ Only depends on M and r, not m

# Binary Systems

Unlike what we were talking about earlier, a smaller satellite orbits a much larger mass (like Earth), a binary system involves two masses orbiting a common center of mass



Features:

- As mentioned, the masses orbit the center of mass (not one orbiting another)
- Same orbital period "T" hence the same angular speed " $\omega$ " ( $\omega = \frac{2\pi}{T}$ )
- Center of mass is closer bigger object

- Force between them =  $\frac{GM_1M_2}{(r_1+r_2)^2}$

- For  $M_1$  :

$$\frac{GM_1M_2}{(r_1+r_2)^2} = M_1\omega^2r_1$$

- For  $M_2$  :

$$\frac{GM_1M_2}{(r_1+r_2)^2} = M_2\omega^2r_2$$

- Ratio of radii and masses :

$$F_1 = F_2 \rightarrow M_1\omega^2r_1 = M_2\omega^2r_2$$

$$\rightarrow M_1r_1 = M_2r_2 \rightarrow \frac{M_1}{M_2} = \frac{r_2}{r_1}$$

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